

Woodbridge Nutrition Platform

A secure school nutrition, family payments, and cafeteria operations pilot for Venderly

Document status	Prepared for	Version
Proposed pilot scope	Venderly & Woodbridge School District	1.0 - August 2026

Purpose

This PRD defines the first demo-ready version of the Woodbridge Nutrition Platform. It is intentionally designed as a safe, end-to-end pilot: realistic workflows, synthetic records, test payments, and a simulated Infinite Campus import. It is not a production meal-claiming system and must not be used with real student records until district approvals, data agreements, and integration decisions are complete.

Product promise

Give guardians clear control of family meal funds, let cafeteria teams move students through the line quickly, and give district staff a reliable, auditable picture of money and meal activity.

Prototype labels required on all product surfaces

PROTOTYPE - SYNTHETIC DATA. This demonstration is not connected to Infinite Campus, PCS, or live payment processing.

1. Product context and objectives

Woodbridge currently uses Infinite Campus as the student information source and sends a one-way roster export to PCS. The proposed product adds a modern nutrition operations layer: household self-service, a fast cafeteria experience, balanced accounts, and district visibility.

Primary objectives

- Demonstrate a polished, credible school nutrition platform to district stakeholders.
- Reduce manual work for balance deposits, corrections, and sibling transfers.
- Protect student privacy through minimal data collection and eligibility confidentiality.
- Provide a foundation that can later integrate with approved district systems.

Success measures for the pilot

Guardian experience	A guardian can view only their household, add a test deposit, review activity, and transfer funds between two linked children.
Cafeteria flow	A cashier can record breakfast/lunch from a student number with a clear confirmation and no visible eligibility category.
Operations	An administrator can find a student, review the full ledger, make an audited adjustment, and export a report.
Trust	Every money movement is traceable; the pilot contains only synthetic data.

Non-goals for the pilot

- Live Infinite Campus, PCS, GoHighLevel, or payment-provider integrations.
- Federal reimbursement claims or replacing PCS as the official meal-counting source.
- Use of real student or guardian information.
- A production offline-payment guarantee.

2. Users, roles, and access boundaries

Access must be enforced by the application backend. Screens may simplify a role's experience, but they cannot be the only protection.

Guardian	Own linked household only	View child balances and history; add funds; split deposits; transfer funds between linked children.
Cashier	Assigned school; POS only	Record meals and sell optional items; cannot browse students, view eligibility, or adjust money.
School staff	Assigned school	Read student details and activity for their school; no unrestricted financial control.
District admin	All assigned schools	Search students, adjust and reallocate funds with reasons, run reports, review records.
Super admin	District-wide configuration	Manage users, school configuration, item/price settings, and audit visibility.

Privacy rules

- A guardian query must always include the authenticated guardian relationship; no open student search.
- Cashiers receive only an operational decision - never free/reduced/paid status.
- Staff access is limited to the district and school boundaries assigned to their role.
- Exports and manual adjustments are audit events.

3. Functional requirements

FR-1 - Student and household directory

- The system shall store a minimal student roster: student number, first/last name, school, grade, and enrollment status.
- The system shall model authorized guardian-to-student relationships.
- The pilot shall seed four schools, approximately 200 students, and realistic multi-child households with differing surnames.

FR-2 - Guardian portal

- Secure sign-in to a household dashboard with one card per linked child.
- Show current balance, low-balance state, and child-specific transaction history.
- Allow a test deposit to one child or split across two or more linked children.
- Allow transfer of available funds among linked children with clear confirmation.

FR-3 - Cafeteria point of service

- Home screen has large Breakfast and Lunch actions.
- Cashier enters a student number using keyboard or numeric keypad.
- System validates student enrollment and school assignment, prevents a duplicate meal event, applies the configured rule, then confirms the result.
- Optional a-la-carte tiles deduct from the available balance and stop a purchase that would go below zero.

FR-4 - Administration

- Search students by number or name within permission scope.
- Show balance, transactions, meal events, guardians, and audit history.
- Support reallocation, refund/adjustment, and correction workflows with a mandatory reason.
- Provide daily meal counts per school, monthly deposits per school, and transaction export.

4. Business rules and decision logic

Meal pricing

Free meal	Record the meal event; charge \$0.00.
Reduced-price meal	Record the meal event; deduct configured reduced price.
Paid meal	Record the meal event; deduct configured full price.
CEP configuration	Default Woodbridge demo behavior: breakfast/lunch at \$0.00 for all; snack balances remain active.
A-la-carte sale	Deduct configured item price; deny when available balance is insufficient.

Financial integrity

- All monetary values are integer cents.
- The ledger is append-only. A correction creates a new offsetting entry; it never edits history.
- A sibling transfer produces a linked debit and credit with a single transfer reference.
- Payment or import events have idempotency keys so retries never double-credit funds.
- Account balance is derived from the ledger; any cached balance is an optimization, not the source of truth.

Meal integrity

- One meal event per student, service date, and meal type is permitted unless a documented override is authorized.
- Price and eligibility decisions occur server-side.
- The product records meal events for operations and demo reporting only unless the district formally authorizes it as an official count source.

5. Proposed system architecture

Recommendation: a modular monolith. One well-separated application with clear modules is safer and faster to deliver than prematurely splitting the pilot into microservices.

React web application	Guardian portal, tablet-ready POS, and administrative console using a shared component system.
Application API	Authentication, authorization, validation, product workflows, reports, and integration adapters.
Domain modules	Directory, household, ledger, POS, meals, reports, import, audit, configuration, and notifications.
PostgreSQL	Transactional records, strict foreign keys and unique constraints, role-scoped data access.
Background jobs	Imports, notification delivery, report generation, retention tasks, and retryable integration events.
Integration adapters	Approved Infinite Campus import, payment events, and optional Venderly/GoHighLevel notification workflows.

External system boundaries

- Infinite Campus remains the authoritative source for student/guardian/school data.
- The first integration approach should be a secure scheduled roster-file import, matching the confirmed current export model.
- The nutrition application owns balances, ledgers, meal events, items, and reports.
- Venderly/GoHighLevel may support communication and workflows, but should not be the system of record for protected nutrition data.
- Payment cards remain with the hosted payment provider; the application consumes verified completion events only.

6. Core data model

Primary entities

District / School	Tenant and permission boundary	Every operational record belongs to a district; school-scoped records require school membership.
Student	Minimal roster record	Unique student number within district; inactive records are retained for history, not deleted.
Guardian / GuardianStudent	Authorized household relationship	Guardian access only exists through a verified relationship.
Account / LedgerEntry	Student balance and money movements	Ledger entry uses integer cents, immutable record, unique idempotency key.
MealEvent	Breakfast/lunch recording	Unique student + service date + meal type prevents accidental duplicate counts.
Item / ItemSale	A-la-carte catalog and sale	Sale stores price at time of purchase for historical accuracy.
AuditLog	Evidence of sensitive action	Actor, action, subject, timestamp, reason, and before/after context.

Data minimization

The required roster fields are student number, first name, last name, school code, grade, and enrollment status. Guardian relationships require only the fields needed for account access and communication. The product must not import race, ethnicity, federal racial designation, gender, or similar fields because these have no nutrition-payment purpose. Date of birth must be excluded unless a future, approved verification design demonstrably requires it.

7. Critical user journeys

Guardian deposit and sibling transfer

1. Sign in	Guardian reaches only their household dashboard.
2. Select action	Guardian chooses Add money or Transfer funds.
3. Choose children and amount	Only linked children are offered; amount is validated.
4. Confirm	For deposits, redirect to hosted test checkout. For transfers, display clear source/destination confirmation.
5. Record	Create immutable ledger records and an audit event.
6. Communicate	Update visible balance immediately and show receipt/confirmation.

Cafeteria meal flow

1. Choose meal	Cashier selects Breakfast or Lunch.
2. Enter number	Student number is entered from keyboard/keypad.
3. Resolve privately	Server validates school, enrollment, duplicate status, and applicable rule.
4. Confirm	Display student name, school/grade if needed, and a neutral "Meal recorded" confirmation.
5. Next student	Auto-reset to entry in roughly two seconds.

Admin correction

- District admin searches within authorized school scope.
- Admin selects a correction or reallocation type and enters a required reason.
- System creates new ledger entries, links related records, and writes an audit event.
- History clearly shows both original and corrective activity.

8. Non-functional requirements

Performance	Normal POS validation target: under one second end-to-end. Guardian/admin views should feel immediate under ordinary district load.
Availability	Pilot operates online. Production readiness requires managed database backups, health checks, monitoring, and incident procedures.
Security	Encryption in transit and at rest, backend authorization, secure sessions, multi-factor staff sign-in, rate limiting, and least-privilege credentials.
Accessibility	Keyboard navigation, visible focus, adequate contrast, large touch targets, labels, and no critical color-only states.
Auditability	Log sensitive access, adjustments, transfers, exports, configuration changes, and integration/import outcomes.
Reliability	Idempotent payment/import processing, transaction boundaries around ledger updates, retryable background work, and alerting for failures.
Privacy	Synthetic data in all demo/test environments; retention and deletion rules before production.

Offline position

Offline serving-line behavior is a production discovery item. The pilot may demonstrate online flow only. A later production design can securely queue encrypted local meal events and synchronize them with idempotency keys once the district defines its outage, device, and reconciliation rules.

9. Integration and import requirements

Infinite Campus roster import

- Accept approved scheduled CSV delivery or another district-approved secure exchange method.
- Validate file type, size, required headers, row counts, school codes, duplicate student numbers, and malformed values before committing changes.
- Present an import result: created, updated, inactive, skipped, failed, and actionable error rows.
- Use upsert behavior and mark missing/enrollment-ended students inactive instead of deleting records.
- Record source, operator/service, timestamp, file checksum/reference, and result in import audit history.

Current known export headers

student.studentNumber	Required
student.firstName / student.lastName	Required
function.SchoolCode	Required
student.grade	Required
student.middleName	Optional; do not display unless needed
student.birthdate	Exclude unless approved verification requires it
student.raceEthnicityFed / student.gender	Do not import

Payment integration

The pilot uses a simulated or hosted test checkout. In production, create a payment intent/checkout session, verify the provider's signed event server-side, then create exactly one ledger credit keyed to the provider event identifier. Never accept a client-side "payment succeeded" message as proof of a deposit.

10. Reports, dashboards, and audit evidence

District dashboard	Meal counts, deposits, adjustments, low-balance trends, and outstanding operational exceptions by school.
Student detail	Current balance, ledger history, meal events, item sales, guardians, and changes.
Daily meal report	Counts by school, meal type, service date, and status.
Monthly deposits report	Deposits, transfer activity, refunds/adjustments, and totals by school.
Transaction export	Authorized downloadable transaction data with filters and export audit record.
Audit log	Actor, action, subject, time, source, reason, and before/after context for sensitive changes.

Reporting guardrails

- Reports must not expose eligibility categories except to explicitly authorized users with an approved operational need.
- Guardian reports contain only the authenticated household.
- Every export is permission-checked and logged.
- Financial reports are calculated from the append-only ledger.

11. Delivery plan and acceptance criteria

1. Foundation	Synthetic seed data, roles, navigation, core schema	Each role sees only its permitted surface and sample data.
2. Guardian portal	Household cards, history, deposits, transfers	A guardian completes deposit simulation and sibling transfer without staff assistance.
3. Ledger	Credits, debits, transfers, adjustments, audit	Every balance change is explainable from ledger history.
4. POS	Meal entry, duplicate guard, optional sales	Cashier records a meal quickly without seeing eligibility.
5. Admin	Search, detail, reallocation, reports	Admin completes a documented correction and exports a report.
6. Import	CSV validation and import report	System accepts approved synthetic roster and produces a clear error/result summary.
7. Demo hardening	Responsive/accessibility polish, labels, script	All workflows run end-to-end with synthetic data and prototype labels.

Pilot demo sequence

- Open on the cafeteria POS: choose Lunch, enter student number, show fast neutral confirmation.
- Switch to guardian portal: show two children, add a test deposit, and transfer funds between siblings.
- Open admin view: review the linked ledger entries, demonstrate an audited correction, and show reports.
- Close with the import/audit view and data-minimization promise.

12. Risks, dependencies, and decisions needed

CEP status	Woodbridge may provide free breakfast/lunch for all students.	Confirm status across all schools for the coming year; retain configurable pricing.
PCS scope	Unclear whether Venderly will replace meal counting or start alongside PCS.	Start parallel / non-claiming unless district formally authorizes a system-of-record change.
Guardian links	Infinite Campus is expected to hold authoritative relationships.	Confirm available guardian export/API fields and approved linking flow.
Payment ownership	Merchant-of-record and settlement flow are not confirmed.	Confirm district/Venderly payment-account model before production.
Data agreement	Student data requires district approval and safeguards.	Use synthetic data now; complete agreement, privacy, security, retention review before live use.
Offline operation	Operational need is recognized but not defined.	Gather cafeteria outage/device/reconciliation requirements before committing a design.

Approval checkpoint

Before production development begins, Venderly and district stakeholders should approve: product scope, meal-counting responsibility, data exchange method, household identity model, payment ownership, privacy/security obligations, and offline expectations.

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